How much of the stem must a terrestrial laser scanner see? Diameter error as a function of circumferential coverage, and what it means for scan design

Author: Angel Naya, Independent Researcher, Berlin, Germany
Corresponding author: Angel Naya, angel.naya@posteo.de

> DRAFT — 14 Oct 2026. References checked against Crossref/Zenodo (`lit/refs_check.md`). Open: Zenodo archive of the code release at submission.

Abstract

Terrestrial laser scanning (TLS) inventories estimate diameter at breast height (DBH) by fitting circles or cylinders to stem points, but occlusion means that each stem is seen only on part of its circumference. Software therefore applies coverage rules — 3DFin, for example, accepts a diameter only when at least 9 of 16 angular sectors (202.5°) contain points — yet how much coverage is actually needed, and whether one threshold suits all estimators, has not been quantified. We combine a Monte Carlo experiment (13 arcs, six estimators, seven perturbation scenarios including non-circular cross-sections, outliers from branches and understorey, and inter-scan registration offsets) with HELIOS++ simulations of three stands (400, 800 and 1200 stems ha^{-1}) scanned from 40 candidate positions, recombined offline into 216 scan configurations (26 712 stem × configuration cases). The visible arc is computed geometrically, independent of point density. For circular stems without outliers, 45–135° of visible arc suffices for a 2 cm RMSE; non-circularity raises this to about 180–202.5° for 40 cm stems, and with 5–15 % outliers only robust estimators reach it (RANSAC from 112.5–135°, Huber 135–270°, Hough 180–225°), while least-squares fits never do, even at full coverage. In the simulated stands, 82 % of stems are within 2 cm at 180–225° visible arc and over 90 % beyond 225° (Kasa and cylinder fits). The 3DFin sector rule corresponds to a median of about 190° geometric arc and separates stems with 92 % from those with 63 % of estimates within 2 cm. Four positions on a grid bring 80–89 % of stems above 180°; four positions clustered near the plot centre fall to 56 % at 1200 stems ha^{-1}. On 456 labelled real stems (SegmentedForests) the Kasa fit needs about 225° for 80 % of stems within 2 cm; with the surrounding points left in, least-squares fits saturate at 55 % even at full coverage, whereas RANSAC reaches 84 %, and 62 % against field DBH on 460 well-covered LAUT trees. A coverage threshold is therefore defensible, but it must be stated per estimator, per expected stem shape and per segmentation quality, and scan positions must be spread rather than added.

Keywords: terrestrial laser scanning; diameter at breast height; occlusion; circle fitting; scan design; HELIOS++; forest inventory

1. Introduction

TLS has become a standard instrument for plot-level forest inventory, yet occlusion remains its main limitation: stems hide one another, and the share of a stem’s

circumference that reaches the scanner depends on stand density, diameter distribution and the placement of scan positions (Abegg et al., 2017; Liang et al., 2018). Single-scan inventories detect markedly fewer stems than multi-scan inventories, especially in dense, multi-layered stands (Liang et al., 2018).

Detection is only half of the problem. A stem that is detected may still be seen on a short arc, and the diameter fitted to a short arc is poorly constrained: algebraic circle fits are known to be biased towards small radii on short, noisy arcs (Kåsa, 1976; Al-Sharadqah and Chernov, 2009; Chernov, 2010). Inventory software reacts with coverage rules. 3DFin (Laino et al., 2024) divides the cross-section into 16 sectors and accepts a diameter only if at least nine contain points, i.e. 202.5°. In our own processing of an open TLS plot of the ICOS site Renon (Chang et al., 2025), this rule rejected the diameter of 29 of 37 detected stems, because the median coverage of four closely spaced scan positions was 180°. On the two Vienna Woods plots of the SegmentedForests data set (Laino et al., 2026), where median coverage was 280–360°, the same software reached 95–100 % precision. Whether the rule was too strict at Renon, and whether it is appropriate for other estimators, is an open question.

Abegg et al. (2017) ray-traced 2740 virtual plots derived from the Swiss national forest inventory and showed that visibility depends on stem number, dominant diameter and the Weibull scale of the diameter distribution, and that evenly distributed scan positions give the best visibility; they did not estimate diameters. Gollob et al. (2019) and Ritter et al. (2020) related scanner position layout and plot size to detection rate and diameter accuracy on real inventory plots, and FORTLS (Molina-Valero et al., 2022) corrects stand estimates for occlusion in single-scan inventories. Li et al. (2021) introduced the cumulative degree of ring closure (CDRC) — the union of the angular sectors of a stem circle that are visible from the scan positions — to optimise scan positions in HELIOS-simulated forests, and noted that accurate DBH estimation requires a CDRC above about 200°. That statement rests on one least-squares estimator, idealised segmentation and simulated stems. How the threshold depends on the estimator, on non-circular stems, on outliers from branches and understorey, and whether it holds for real bark and field-measured DBH, has to our knowledge not been quantified, nor has it been related to the sector rules used in inventory software.

We ask: (i) how does the DBH error of common estimators depend on visible arc, and how is this modified by non-circular stems, outliers and registration offsets; (ii) does a sector-based rule such as 3DFin’s correspond to a meaningful coverage threshold; and (iii) which scan designs achieve that coverage in stands of different density?

2. Methods

2.1 Estimators

All estimators operate on the points of one stem in the band 1.05–1.55 m (cylinder: 1.0–1.6 m):

- **Kasa**: algebraic least-squares circle (Kåsa, 1976).
- **Geometric**: least-squares circle minimising orthogonal distances (Gauss–Newton, bounded centre ±0.3 m and radius 4–75 cm; Kasa start).

- **RANSAC**: three-point circle hypotheses (400 iterations, 12 mm inlier tolerance); a hypothesis is admitted only if its inliers span at least 90°; final Kasa fit on the inliers (Fischler and Bolles, 1981).
- **Cylinder**: circle fit in the plane normal to an axis whose tilt is optimised (Nelder–Mead, ≤ 15°), to remove the elliptical bias of tilted stems.
- **Hough**: circle Hough transform with normal-directed voting; normals from local PCA (k = 12) after 1 cm thinning; 5 mm accumulator, 3 × 3 smoothing, radius coarse-to-fine (1 cm, 2 mm).
- **Huber-robust**: geometric fit with Huber loss (scale 1 cm), Kasa start.

Implementations: `scripts/dbh_methods.py`, `scripts/arc_error_mc_real.py` (repository, Section *Code availability*). The 3DFin acceptance rule is evaluated as the number of occupied 22.5° sectors around the stem centre in a 10 cm slice at 1.3 m.

2.2 Monte Carlo experiment

A stem of DBH 20 or 40 cm is observed on one contiguous arc $\theta \in \{45, 67.5, \ldots, 360°\}$ with N = 200 or 1000 points uniformly distributed over the arc and a 0.6 m height band, radial Gaussian noise σ = 5 mm (2 and 10 mm in the ideal case). Seven scenarios: *ideal*; *non-circular* (elliptical cross-section, axis ratio U(0.85, 1), random orientation; true DBH = perimeter/π, tape definition); *5 % / 15 % outliers* (points 2–30 cm outside the surface, as from twigs, foliage or understorey in front of the stem); *5 / 10 mm registration offset* (the second half of arcs > 90° shifted, mimicking a second scan position); *combined* (non-circular + 5 % outliers + 5 mm offset). 100 replicates per cell (300 for the ideal grid). The Cramér–Rao lower bound for the radius under the ideal model is reported as a reference for the best achievable precision. Code: `scripts/arc_error_mc.py`, `scripts/arc_error_mc_real.py`.

2.3 HELIOS++ simulation

Three 40 × 40 m core plots with a 10 m buffer were generated (`scripts/paperC_scene.py`): S1a 400 stems ha^{-1} (DBH 38 ± 11 cm), S1b 800 ha^{-1} (30 ± 9 cm), S1c 1200 ha^{-1} (24 ± 7 cm); random positions with a minimum gap of 10 cm. Stems are 4 m elliptical cylinders (axis ratio U(0.85, 1), 32 segments, rings every 0.25 m with ±3 mm bark roughness); crowns were omitted because only the breast-height cross-section is analysed. Scans were simulated with HELIOS++ 2.2.2 (Winiwarter et al., 2022) using the RIEGL VZ-400 model (5 mm range accuracy, 0.3 mrad divergence), tripod height 1.5 m, 0.06° angular resolution, vertical field −30° to +5°. Each scene has a pool of 40 positions (5 × 5 grid at 8 m spacing plus 15 random); each position is simulated once and the configurations are assembled offline: n ∈ {1, 2, 3, 4, 6, 9} positions as (a) a spread grid subset (greedy max–min from the centre), (b) 10 random draws from the pool, (c) the n positions closest to the plot centre (“clustered”, as at Renon). Stem points are assigned by the HELIOS object identifier, i.e. segmentation is perfect, isolating coverage and fitting from detection.

Visible arc. Occupied-sector counts and fine angular bins both depend on point density. We therefore compute the geometrically visible arc: for 360 points on the true elliptical perimeter at 1.3 m, a point counts as visible from a position if it faces it (cosine of incidence > 0.1) and the line of sight does not intersect another stem; the

union over the positions of a configuration gives the visible arc (`scripts/paperC_eval.py`). This is the CDRC of Li et al. (2021), here evaluated on the true elliptical perimeter.

2.4 Metrics

Per estimator and arc class: bias, RMSE, share of stems within ±2 cm (failures counted as outside), failure rate. A *coverage threshold* is the smallest arc from which RMSE ≤ 2 cm, |bias| ≤ 1 cm and failure ≤ 10 % hold for all larger arcs. The 2 cm level corresponds to the DBH accuracy of 1–2 cm that the international TLS benchmark found achievable (Liang et al., 2018).

3. Results

3.1 Monte Carlo

Under the ideal model all least-squares estimators approach the Cramér–Rao bound for arcs ≥ 135°. On short arcs the Kasa, RANSAC and cylinder fits are biased low (DBH 40 cm, N = 200: −2.4 cm at 90°, −7.1 cm at 67.5°), whereas the geometric fit stays unbiased (RMSE 0.8 cm at 90°); coverage thresholds are 45–135° depending on estimator, noise and N (Fig. 1a).

Non-circularity shifts the thresholds of all estimators to about 180–202.5° for DBH 40 cm and to 90–135° for DBH 20 cm: a partial view of an ellipse yields a systematically wrong circle, an error that grows with diameter (Fig. 1b). Registration offsets of 5–10 mm set an RMSE floor of 0.5–0.9 cm but barely change thresholds.

Outliers dominate everything else (Fig. 1c). With 5–15 % outliers, Kasa, geometric and cylinder fits never meet the threshold, even at 360° (RMSE 2–7 cm). RANSAC meets it from 112.5–135°, the Huber-robust fit only from 135–270° (it fails on short arcs because its Kasa starting value is already biased), Hough from 180–225°. In the combined scenario only RANSAC (135–180°), the Huber fit (180–225°) and, partly, Hough (225–315°) reach the threshold.

3.2 HELIOS++: error versus visible arc

Pooled over the three stands (26 712 cases; 56, 130 and 185 core stems), the RMSE of Kasa, cylinder and RANSAC falls from 9–17 cm below 90° to 1.6–1.7 cm at 180–225° and 0.5–0.9 cm beyond 315°; Hough remains worst throughout (Fig. 2a). The share of stems within 2 cm rises from 45–56 % at 90–135° to 82 % at 180–225°, 93 % at 225–270° and 98–99 % beyond 270° for the Kasa and cylinder fits; RANSAC stays slightly lower beyond 225° (89–94 %) (Fig. 2b). Thicker stems need more coverage: at 180–225° RMSE is 2.0 cm in S1a (mean DBH 37 cm) versus 1.5–1.7 cm in S1b and S1c, consistent with the non-circularity effect in Section 3.1. A single scan position sees at most 172° of a stem (median 164°).

3.3 Sector rule and scan design

The 3DFin rule of ≥ 9/16 sectors corresponds to a median geometric arc of 190° at exactly nine sectors (Fig. 4); 7.2 % of accepted stems have less than 180° visible. Used as an acceptance filter, the rule separates stems with 92 % (Kasa) of estimates within 2

cm from those with 63 % — it sits close to the arc at which errors become small in these simulations.

Coverage depends more on the spread than on the number of positions (Fig. 3). Four grid positions bring 89 %, 82 % and 80 % of stems above 180° in the three stands; six bring 96–98 %. Four positions clustered near the centre reach 89 %, 71 % and 56 %, i.e. the penalty of clustering grows with stand density. Random placement performs close to the grid.

3.4 Real point clouds

Data. SegmentedForests (Laino et al., 2026; MIT): 12 of 14 TLS plots with point-wise semantic labels (plot 08 omitted for size, plot 09 failed to download); stem points (class 3) at 1.0–1.6 m above ground were clustered, and 456 stems with ≥ 330° observed coverage (1° bins) and ≥ 300 points served as reference; their reference DBH is the median of the four least-squares/RANSAC estimates on all points (no field DBH is available). LAUT (Gollob et al., 2020a, b; CC BY 4.0): 17 Austrian inventory plots scanned with a Faro Focus3D X330, 1 908 trees with field DBH (median 15.6 cm) and field positions; stem points are all points within DBH/2 + 25 cm of the field position, i.e. without segmentation. Field positions and fitted centres agree to 4 cm (median). Coverage was reduced by retaining one contiguous arc at random orientation (5 repetitions per stem and arc). Breast-height bands were extracted with `scripts/ext_band.py` in a GitHub Actions workflow (`.github/workflows/fetch-data.yml`).

Stem points only (SegmentedForests, Fig. 5a). The Kasa fit reaches 56 % of stems within 2 cm at 180°, 73 % at 202.5°, 82 % at 225° and 91 % at 270°; its RMSE falls from 6.1 cm at 135° to 2.9 cm at 202.5° and 1.5 cm at 270°, with a negative bias on short arcs (−4.4 cm at 135°, −1.0 cm at 202.5°) as predicted by the Monte Carlo experiment. On these thinner stems (median DBH 16 cm), real bark shifts the 80 % point from about 180° (HELIOS) to about 225°. RANSAC with a 12 mm inlier tolerance is weaker on clean real stems (88 % at 360°), because rough bark excludes valid points.

With surrounding points (Fig. 5b, c). When branches, understorey and unlabelled points within 30 cm (SegmentedForests) or 25 cm (LAUT) remain, the least-squares fits saturate: at full coverage only 55 % (Kasa) and 64 % (geometric) of SegmentedForests stems are within 2 cm, and 35–38 % of LAUT trees against field DBH. RANSAC rises with coverage to 84 % (SegmentedForests) and 62 % (LAUT) and is the best estimator from about 112° on, reproducing the outlier scenario of Section 3.1 on real data. The lower LAUT values include field-measurement error and the absence of any segmentation.

4. Discussion

Relation to Li et al. (2021). Their 200° CDRC guideline is confirmed for least-squares fits of clean stem points (HELIOS: 82 % of stems within 2 cm at 180–225°), but real bark and thin stems shift the 80 % point to about 225°, and with surrounding points no CDRC is sufficient for least-squares fits.

A single threshold does not fit all estimators. Without outliers, least-squares fits need far less than 202.5°; with non-circular stems they need about 180–200°; with

outliers they never become reliable. The threshold is therefore a property of the estimator and of the expected cross-section, not of the data alone. The 3DFin rule is well placed for least-squares fitting of moderately non-circular stems after good segmentation, and conservative for a robust estimator on round stems.

Renon revisited. With four closely spaced positions, the Renon plot reproduces the "clustered" configuration. Our simulations predict that many of its stems lie below 180° visible arc and that diameters fitted there carry several centimetres of error — 3DFin's refusal was therefore justified, and estimates that our own pipeline produced for those stems should be flagged rather than reported.

Scan design. In line with Abegg et al. (2017), clustering wastes positions: at 1200 stems ha^{-1}, four clustered positions (56 % of stems ≥ 180°) perform like three grid positions (58 %). A tentative recommendation, to be refined with stand-specific simulations, is a spread grid of four to six positions per 0.16 ha (25–38 ha^{-1}) at 400–1200 stems ha^{-1}, which brought 80–98 % of stems above 180° here.

Limitations. In HELIOS: perfect segmentation (no detection errors), no branches or crowns at breast height, flat terrain, one scanner model, idealised outlier and registration models. Range-dependent incidence effects and beam divergence are simulated by HELIOS but not varied systematically. The Hough implementation (normal voting, 5 mm accumulator) performed worst throughout and may understate what tuned Hough variants achieve. On SegmentedForests the reference DBH is itself a fit to well-covered points, not a field measurement.

5. Conclusions

Coverage rules for TLS diameters should be reported per estimator and per segmentation quality. With clean stem points, least-squares fits need about 180° (simulation) to 225° (real bark) of visible arc for most stems to be within 2 cm, and 3DFin's 202.5° sector rule sits in this range. Without clean segmentation, no amount of coverage rescues least-squares fits; robust estimators such as RANSAC with an arc constraint are then the main lever. Scan positions should be spread rather than clustered, especially in dense stands.

Code availability

- Software: For3Dsuite, https://github.com/anayana/For3Dsuite (GPL-3.0-or-later). Scripts: `arc_error_mc.py`, `arc_error_mc_real.py`, `paperC_scene.py`, `paperC_run.py`, `paperC_pack.py`, `paperC_eval.py`, `paperC_figures.py`, `dbh_methods.py`.
- HELIOS++ 2.2.2 (conda-forge).
- [TODO] Archive release on Zenodo.

Data availability

Result tables (`Paper_ODT/Forst_Benchmark_Paper/results/`) and figure sources are in the repository; simulated scenes are reproducible from the scripts and seeds. Third-party data: Renon/ICOS IT-Ren (Chang et al., 2025; CC BY 4.0); SegmentedForests (Laino et

al., 2026; MIT); LAUT (Gollob et al., 2020b; CC BY 4.0). Derived breast-height bands of SegmentedForests and LAUT are in `extdata/`.

Declaration of competing interest

The author declares no competing interests.

Figures

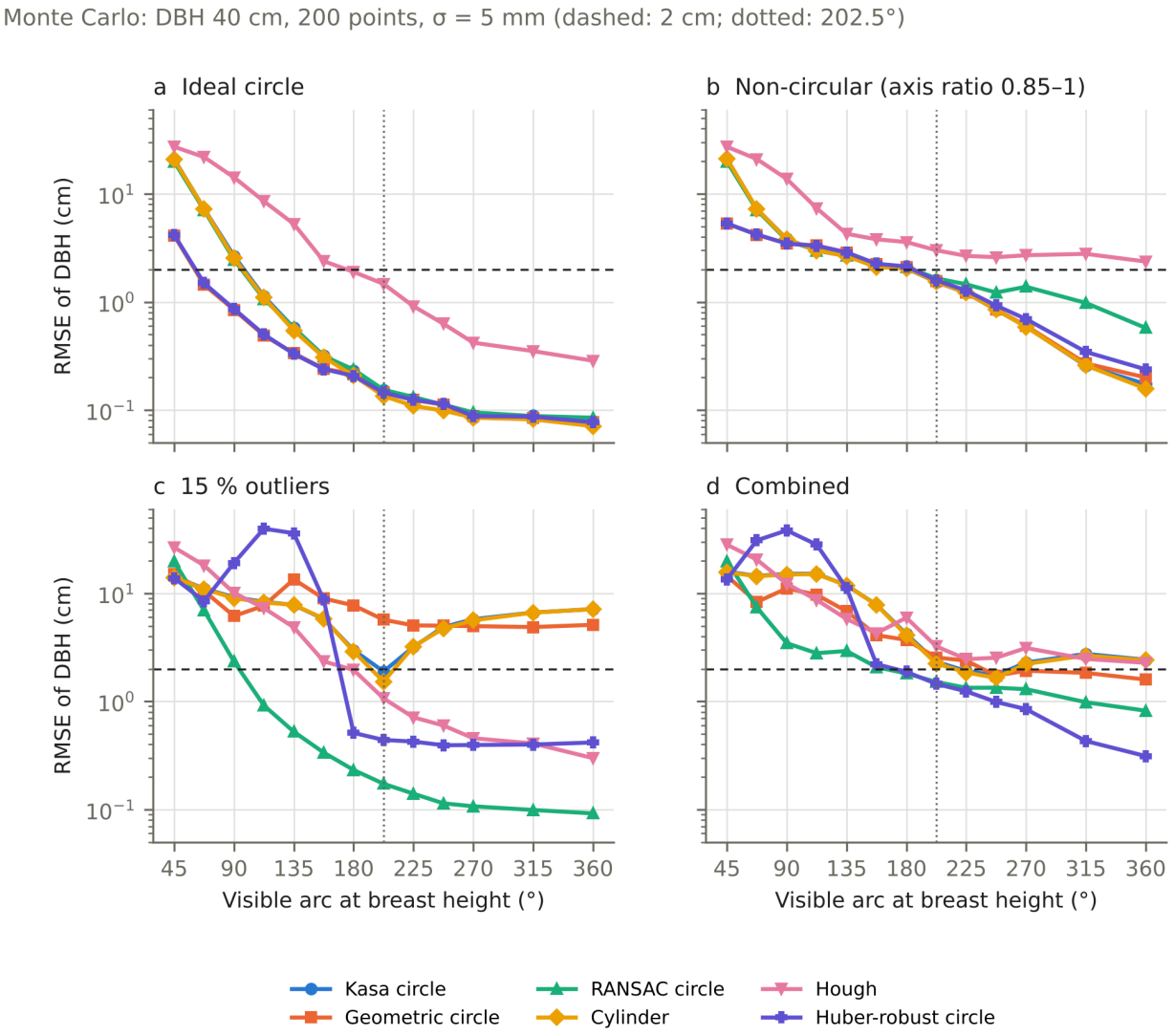


Figure 1. Monte Carlo experiment: RMSE of DBH versus visible arc for six estimators (DBH 40 cm, 200 points, σ = 5 mm). (a) ideal circle, (b) non-circular cross-section (axis ratio 0.85–1), (c) 15 % outliers, (d) combined perturbations. Dashed line: 2 cm; dotted line: 202.5° (3DFin sector rule).

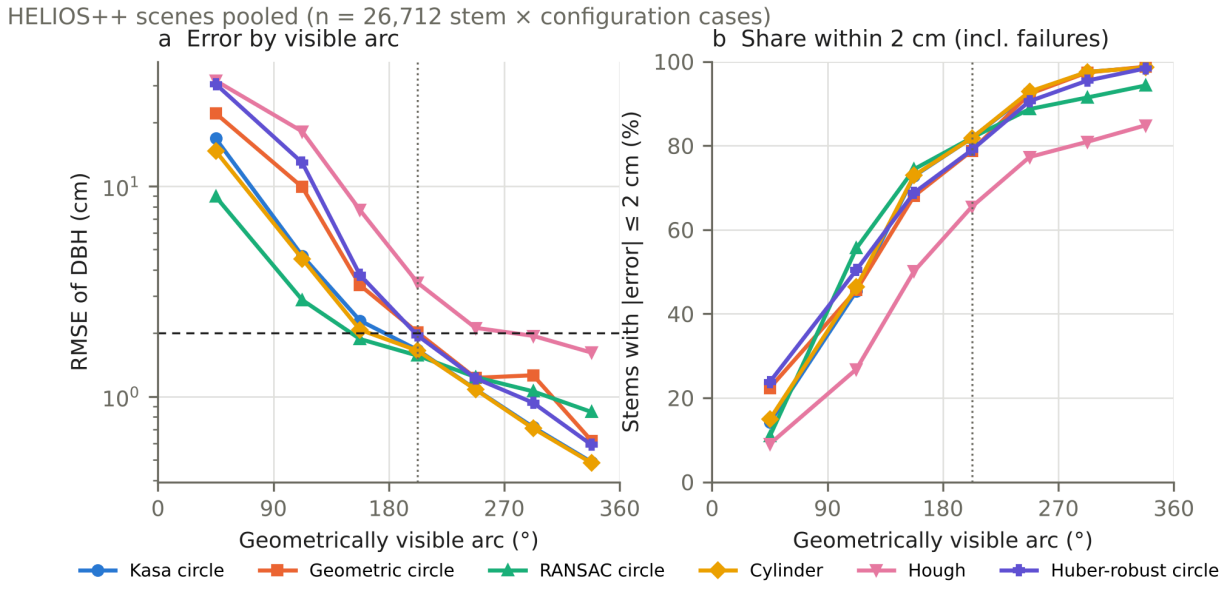


Figure 2. HELIOS++ simulations, three stands pooled (26 712 stem × configuration cases). (a) RMSE of DBH and (b) share of stems within ±2 cm (failures counted as outside) versus geometrically visible arc.

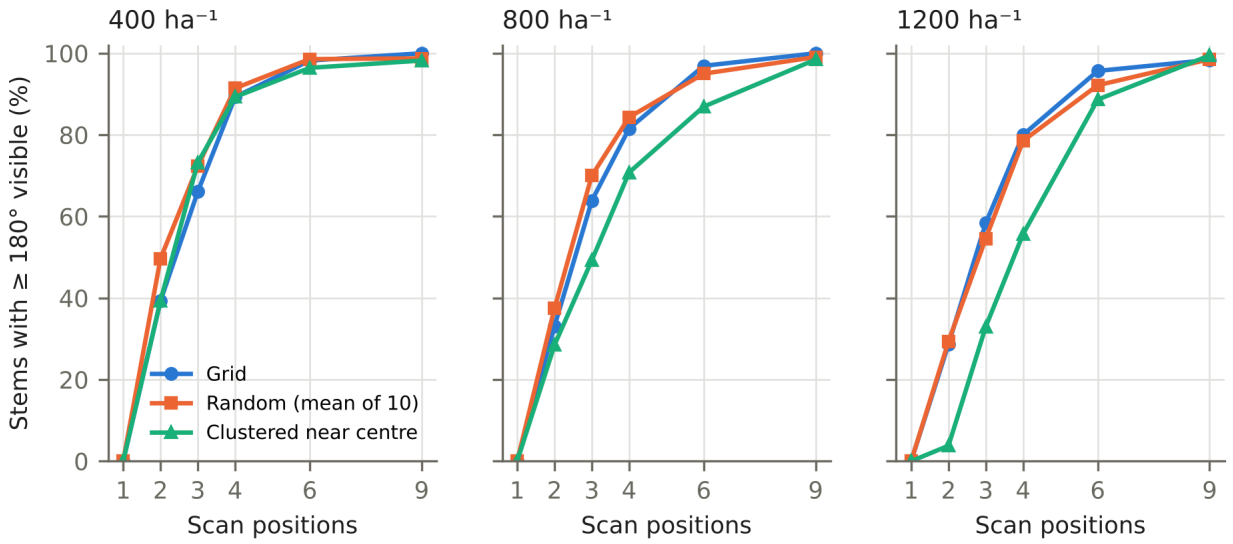


Figure 3. Share of stems with ≥ 180° visible arc versus number of scan positions for grid, random (mean of 10 draws) and clustered placement in stands of 400, 800 and 1200 stems ha⁻¹.

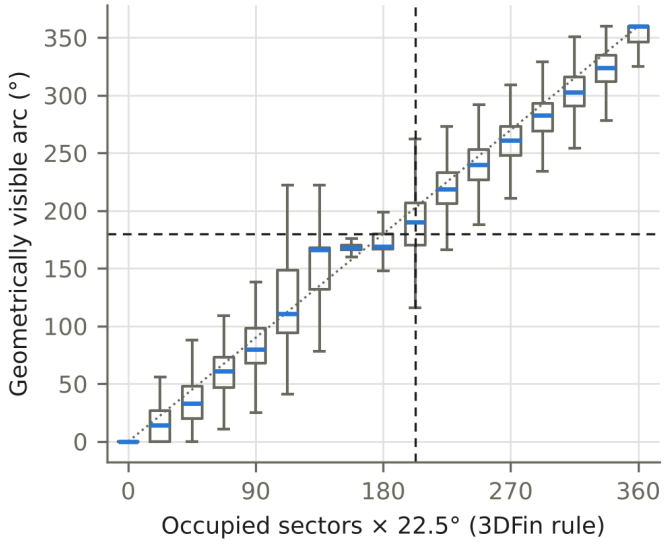


Figure 4. Occupied 22.5° sectors (3DFin rule) versus geometrically visible arc in the HELIOS++ simulations; boxes show quartiles, whiskers 1.5 IQR. Dashed lines: 202.5° sector rule and 180° visible arc.

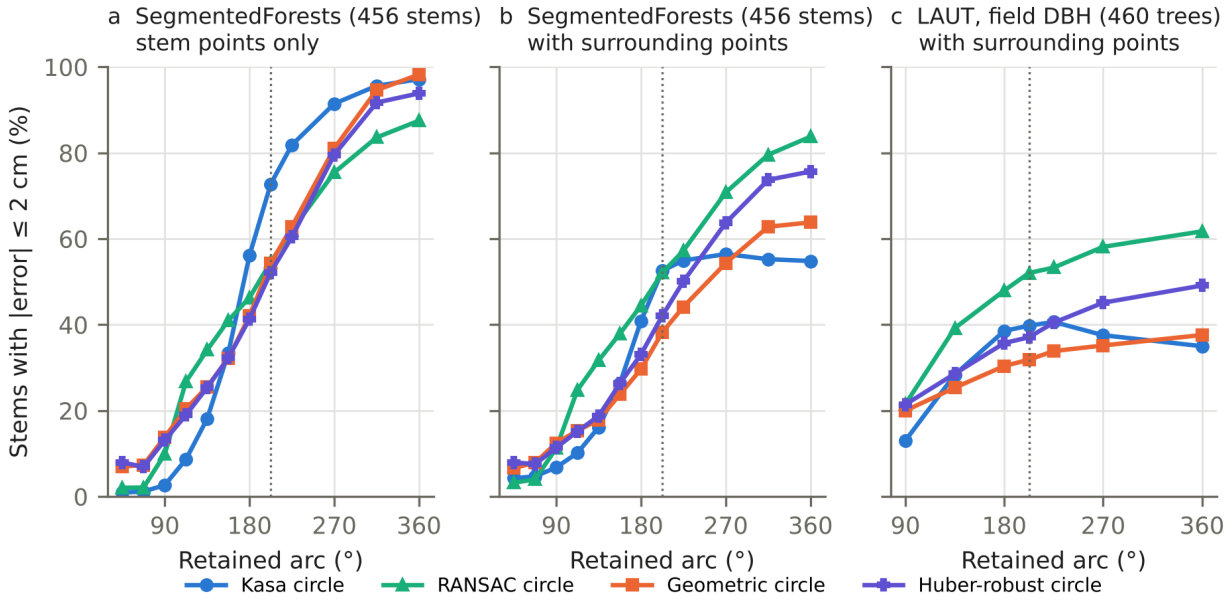


Figure 5. Real point clouds, share of stems within ±2 cm versus retained arc. (a) SegmentedForests, stem points only (456 stems, reference: fit on full coverage); (b) the same stems with surrounding points within 30 cm; (c) LAUT, 460 well-covered trees against field DBH, with surrounding points within 25 cm.

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